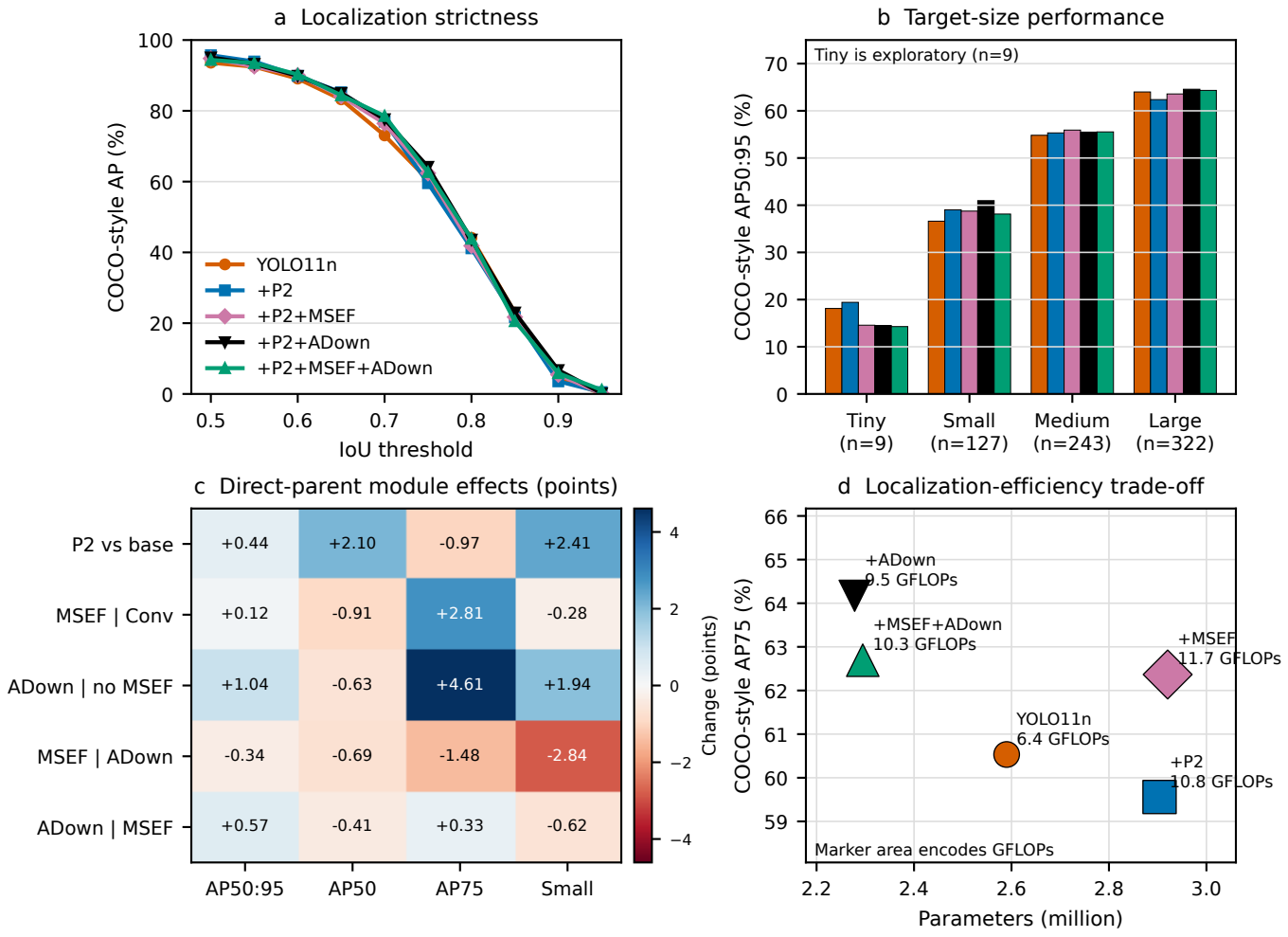
a Localization strictness
COCO-style AP (%)
IoU threshold
YOLO11n
+P2
+P2+MSEF
+P2+ADown
+P2+MSEF+ADown
b Target-size performance
Tiny is exploratory (n=9)
COCO-style AP50:95 (%)
Tiny (n=9)
Small (n=127)
Medium (n=243)
Large (n=322)
c Direct-parent module effects (points)
P2 vs base
MSEF | Conv
ADown | no MSEF
MSEF | ADown
ADown | MSEF
+0.44
+2.10
-0.97
+2.41
+0.12
-0.91
+2.81
-0.28
+1.04
-0.63
+4.61
+1.94
-0.34
-0.69
-1.48
-2.84
+0.57
-0.41
+0.33
-0.62
AP50:95
AP50
AP75
Small
Change (points)
d Localization-efficiency trade-off
+ADown
0.5 GFLOPs
+MSEF+ADown
10.3 GFLOPs
+MSEF
11.7 GFLOPs
YOLO11n
6.4 GFLOPs
+P2
10.8 GFLOPs
Marker area encodes GFLOPs
COCO-style AP75 (%)
Parameters (million)